

C++ Quiz — Output Prediction

30 Questions | 3 Topics | Read the code and write the output

TOPIC 1: Built-in Library — string & cstring

Q1.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Ahmad";
    cout << s.length() << endl;
    cout << s.size() << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: 5 | 5

Hint: length() and size() both return number of characters in the string.

Q2.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Hello World";
    cout << s.substr(6, 5) << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: World

Hint: substr(start, length) — starts at index 6, takes 5 characters.

Q3.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Programming";
    cout << s.find("gram") << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: 3

Hint: find() returns starting index. 'gram' starts at index 3 in 'Programming'.

Q4.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string a = "Hello";
    string b = " World";
    string c = a + b;
    cout << c << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Hello World

Hint: The + operator concatenates two strings.

Q5.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Pakistan";
    s.replace(0, 3, "Hin");
    cout << s << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Hindustan

Hint: replace(pos, len, newStr) — replaces 3 chars from index 0 with 'Hin'.

Q6.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
    char a[] = "Hello";
    char b[] = " C++";
    strcat(a, b);
    cout << a << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Hello C++

Hint: strcat() appends string b to the end of string a.

Q7.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
    char s[] = "Compiler";
    cout << strlen(s) << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: 8

Hint: strlen() counts characters excluding the null terminator. 'Compiler' = 8 chars.

Q8.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
    char a[] = "Apple";
    char b[] = "Apple";
    cout << strcmp(a, b) << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: 0

Hint: strcmp() returns 0 when both strings are equal.

Q9.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
    char src[] = "Lahore";
    char dest[20];
    strcpy(dest, src);
    cout << dest << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Lahore

Hint: strcpy() copies the source string into the destination array.

Q10.

What is the output of the following code?

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Hello Ahmad";
    s.erase(5, 6);
    cout << s << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Hello

Hint: erase(pos, len) removes 6 characters starting from index 5.

TOPIC 2: Character String

Q11.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char name[] = "Ahmad";
    cout << name << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Ahmad

Hint: Printing a char array prints the string up to the null terminator.

Q12.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char s[] = "Hello";
    cout << s[1] << endl;
    cout << s[4] << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: e | o

Hint: s[1] = 'e', s[4] = 'o' (0-indexed).

Q13.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char s[] = "ABCDE";
    for(int i=0; s[i] != '\0'; i++){
        cout << s[i];
    }
    cout << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: ABCDE

Hint: Loop runs until null terminator '\0' is reached.

Q14.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char s[] = "Code";
s[0] = 'M';
cout << s << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Mode

Hint: Character strings stored in arrays can be modified by index.

Q15.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
char s[] = "Karachi";
cout << strlen(s) << endl;
cout << sizeof(s) << endl;
return 0;
}
```

Your Answer: _____

Correct Output: 7 | 8

Hint: strlen = 7 (no null), sizeof = 8 (includes null terminator '\0').

Q16.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char s[] = "Hello";
int i = 0;
while(s[i]){
cout << s[i];
i++;
}
return 0;
}
```

Your Answer: _____

Correct Output: Hello

Hint: Null character is falsy (0), so the while loop stops at the end of string.

Q17.

What is the output of the following code?

```
#include<iostream>
#include<cctype>
using namespace std;
int main(){
char s[] = "hello";
for(int i=0; s[i]; i++)
s[i] = toupper(s[i]);
cout << s << endl;
return 0;
}
```

Your Answer: _____

Correct Output: HELLO

Hint: toupper() converts each character to its uppercase equivalent.

Q18.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char s[] = {'A','h','m','a','d','\0'};
cout << s << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Ahmad

Hint: A char array with explicit null terminator '\0' prints as a string.

Q19.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char s[] = "Pakistan";
cout << s+4 << endl;
return 0;
}
```

Your Answer: _____

Correct Output: stan

Hint: s+4 is a pointer to index 4, so it prints from 'stan' onward.

Q20.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char s[] = "Quiz";
s[2] = 'i';
s[3] = 'z';
cout << s << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Quiz

Hint: s[2] was already 'i' and s[3] was already 'z', so no change occurs.

TOPIC 3: Character Array

Q21.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char arr[5] = {'A','B','C','D','E'};
    for(int i=0; i<5; i++)
        cout << arr[i];
    cout << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: ABCDE

Hint: Each character is printed individually using the loop.

Q22.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char arr[] = {'X','Y','Z'};
    cout << sizeof(arr) << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: 3

Hint: No null terminator — only 3 chars declared, so sizeof = 3.

Q23.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
    char arr[3][4] = {"Cat","Dog","Cow"};
    cout << arr[0] << endl;
    cout << arr[2] << endl;
    return 0;
}
```

Your Answer: _____

Correct Output: Cat | Cow

Hint: 2D char array: arr[0] = first string, arr[2] = third string.

Q24.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char arr[3][6] = {"Apple", "Mango", "Grape"};
for(int i=0; i<3; i++)
cout << arr[i] << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Apple | Mango | Grape

Hint: Each row of the 2D char array is a separate string.

Q25.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char arr[] = "Hello";
cout << arr[0] << arr[4] << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Ho

Hint: arr[0] = 'H', arr[4] = 'o'.

Q26.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char arr[10] = "CS";
cout << sizeof(arr) << endl;
return 0;
}
```

Your Answer: _____

Correct Output: 10

Hint: sizeof gives the declared array size (10), not the string length (2).

Q27.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char a[5], b[5];
a[0]='H'; a[1]='i'; a[2]='\0';
b[0]='B'; b[1]='y'; b[2]='e'; b[3]='\0';
cout << a << " " << b << endl;
return 0;
}
```

Your Answer: _____

Correct Output: Hi Bye

Hint: Each array prints as string until null terminator is found.

Q28.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char arr[5] = {0};
arr[0] = 'O';
arr[1] = 'K';
cout << arr << endl;
return 0;
}
```

Your Answer: _____

Correct Output: OK

Hint: {0} initializes all elements to null. Only 'O' and 'K' are set.

Q29.

What is the output of the following code?

```
#include<iostream>
#include<cstring>
using namespace std;
int main(){
char arr[3][10] = {"Red", "Green", "Blue"};
cout << strlen(arr[1]) << endl;
return 0;
}
```

Your Answer: _____

Correct Output: 5

Hint: strlen(arr[1]) = strlen("Green") = 5.

Q30.

What is the output of the following code?

```
#include<iostream>
using namespace std;
int main(){
char arr[3][4] = {"Cat", "Dog", "Cow"};
cout << arr[1][1] << endl;
return 0;
}
```

Your Answer: _____

Correct Output: o

Hint: arr[1] = "Dog", arr[1][1] = 'o' (index 1 of 'Dog').